Formula Zynerji

A Blockchain-Based, Mechanism-Design Redesign of Formula 1 for Meritocracy

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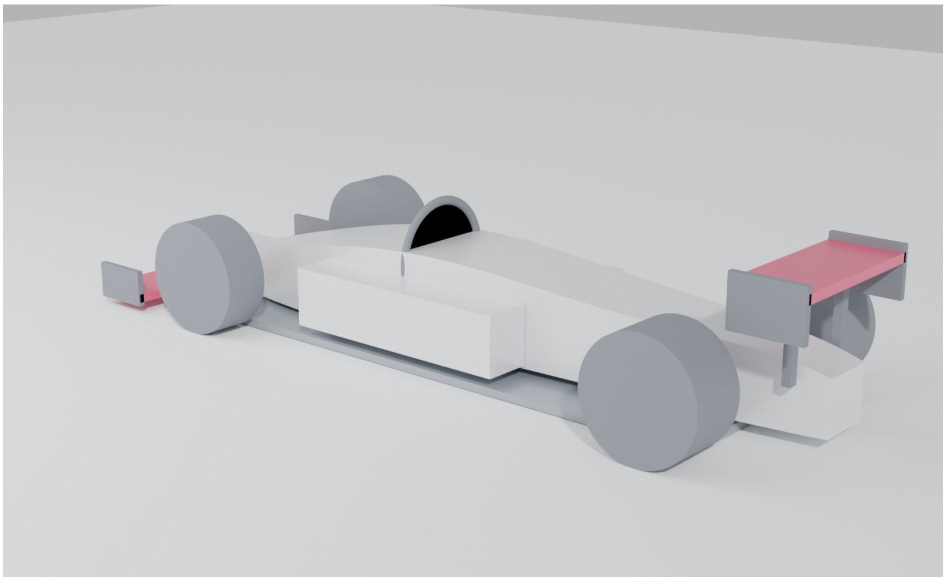

Concept render — 2.5 L two-stroke uniflow diesel, era-kitbash chassis (Blender Cycles, path-traced).

Abstract

Formula Zynerji re-derives the Formula 1 rulebook from a single objective — **meritocracy** — using game theory and seventy years of how teams have actually behaved. Its spine is a **blockchain**: every part that runs on a car must have its full design (CAD/FEA/CFD) uploaded to an immutable ledger, and only on-chain parts are legal. Forced disclosure converts R&D from a private secret into a shared club good, which collapses the spending war and lets the field *self-balance by copying* — with no development tokens, no auction, and no Balance of Performance. A mandatory minimum-spend floor forces continuous innovation; the resulting shared, race-validated engineering corpus is sold to outside industry to fund the sport and the prize money. The car is a deliberate kitbash of historical F1 eras around a distinctive 2.5 L two-stroke uniflow diesel. This paper presents the thesis, the mechanisms (catalogued M1–M13), the car, the competition format, and the economic model that pins the two keystone numbers.

1 Thesis: the championship as a measurement instrument

A season is an *estimator* of an unobservable quantity — true competitive merit (car + team + driver). The standings are the estimate; meritocracy is the program of minimising the gap between them:

$$\text{Standings} = \text{Merit} + \text{Luck} + \text{Budget-bias} + \text{Gaming-bias}.$$

Every rule is an intervention on one of those error terms, and is judged by one test: does it make the finishing order track merit more closely? Formula 1's instinct has been to remove danger and cost by *banning* innovations (flat floors 1983, the turbo ban 1989, the electronic-aid ban 1994). Formula Zynerji keeps the engineering freedom of the 1966–1982 open era but contains its two killers — danger and runaway cost — with a tight *box* (a non-negotiable safety floor, a single energy/power pinch-point, and a self-balancing economy) rather than prescriptive bans. The design razors and the full mechanism catalogue (M1–M13) live in the project's `mechanism-design` document.

2 The spine: forced on-chain disclosure (M10/M11)

Every run part's complete design record is committed to an immutable, timestamped chain by the start of each event; **only on-chain parts are legal**, so scrutineering reduces to a hash check and a secret or back-dated part is structurally impossible. The economic consequence is the point: an innovation, once raced, is visible and copyable, so the value of any one advantage is capped at the head-start it buys. No team will spend millions for an edge that arms its rivals within a weekend — equilibrium R&D spend collapses, and the contest shifts from *hiding the biggest secret* to *innovating fastest and integrating best*. The innovator's reward is the transient head-start (you are fast first, and score points, before rivals can manufacture a copy) plus reputational recognition on a published Innovation Index. There are no tokens and no bounty.

The mechanics are specified to the record level (`blockchain-spec.md`). Crucially, scrutineering is a *two-clause* test, because a manufactured part can never be bit-identical to its CAD: (i) a **cryptographic** check that the fitted part cites an on-chain design whose canonical hash is unaltered and whose timestamp precedes the upload deadline — which makes a secret or back-dated part impossible; and (ii) a **metrology** check that the physical part conforms to that disclosed design within tolerance. Cheating is thereby reduced from "hide a secret part" (impossible) to "manufacture out of tolerance" (a conventional, detectable problem).

3 The self-balancing economy and its revenue

Because forced disclosure makes copying free, the field balances itself: any advantage is copied away (the field compresses), the championship pulls every team toward the best on-chain parts, and re-leading demands fresh innovation. This removed the need for the artificial balancing machinery of earlier drafts — development tokens, a development auction, and a success handicap are all **deleted**. The only economic levers are a hard, money-neutral **cap** and a mandatory minimum-spend **floor**. The floor is the keystone: a token-free copy economy has exactly one bad equilibrium — everyone copies, nobody innovates — and the floor forecloses it by forcing every team to do original development.

The shared corpus is also a **product**. Forced disclosure makes the chain a continuously-growing, cross-validated, race-proven engineering dataset, sold through tiered access: competitors get it free (disclosure is how they compete); journalists pay a media tier; and **non-competing industry** — aerospace, defence, automotive and energy firms such as Boeing and Lockheed — license the deep CAD/FEA/CFD, materials and control data for their own R&D at a premium. This is the largest stream, and it funds the series' operations, the **prize money** for the four championships, and a redistribution toward smaller teams. Richer innovation makes the corpus more valuable, which grows the prize fund — a further reason to innovate.

4 The car: an era kitbash

The car deliberately ports specific historical F1 regulation sets: **2026** wings, floor, tyres and safety; **2021** hydraulically-interconnected suspension; a **2013** 7-speed gearbox; and **2008** chassis dimensions and body rules. A 3300 mm wheelbase, 1800 mm width and ∼605 kg minimum mass give a planted but ferociously powerful machine. Aero is disciplined not by ever-more-prescriptive geometry but by *dirty-air externality pricing* (M4): each car's wake is measured, and a clean-wake car is granted a larger design-stage downforce allowance — so close racing and overtaking happen on merit, and there is no DRS. This is a concrete schedule, not a slogan: an illustrative calibration (`aero_dirty_air.py`) shows the price moving the privately-optimal design from a ∼45% follower-downforce loss (2021-generation, effectively unraceable) to ∼16% (close to the 2022 ground-effect target) at only ∼15% less realised downforce — raceability bought by private incentive rather than by Balance of Performance.

5 The powertrain: a 2.5 L two-stroke uniflow diesel

The engine is a 2.5 litre inline-five, two-stroke, uniflow-scavenged compression-ignition (diesel) unit on a JP-8-class synthetic e-kerosene, force-fed by a mechanical screw (Lysholm) supercharger, with no traction hybrid. Uniflow scavenging — fresh charge in through liner ports at the bottom, exhaust out through poppet valves in the head — is the most efficient two-stroke method and underpins the high BMEP. At 26 bar BMEP and 7000 rpm it makes ∼758 kW (∼1015 hp) and ∼1035 N m; at the 605 kg minimum that is ∼1.25 kW/kg, turbo-era-F1 power-to-weight. Exhaust tuning is by an electronically-controlled variable-length manifold (set-and-run): an automated *engine* optimisation, not a driving aid. These figures are derived, not asserted: a boost–energy balance (`engine_cycle.py`) reproduces them from a pressure-ratio ≈3.3 intercooled screw blower at λ1.3 and 44% brake thermal efficiency, which also fixes a ∼190 g/kWh BSFC, a 20 m/s mean piston speed at the ceiling (below F1's ∼25), and an 86 mm square bore/stroke — and the 80 kg fuel load then supports ∼280 kW average over a 90 minute race.

6 Driver-skill primacy and the vectoring differential

The governing electronics principle is *automate the engine, never the driving*: live manual driver inputs are encouraged; automated engine-output optimisation is permitted but locked before the race; automated car-control aids (traction control, ABS, closed-loop vectoring) are banned. The rear differential is a **driver-vectored** twin wet-clutch unit (GKN Twinster-type): two steering-wheel triggers set the percentage lockup of each side's clutch in real time. It is legal precisely because it is *open-loop* — the trigger-to-clamp map is fixed, declared and on-chain, and the safety ECU adds no sensor feedback — which is what distinguishes it from banned automated vectoring.

7 The competition: a merit estimator

The competition is tuned for signal-to-noise. The points curve scores *all* classified finishers on a convex 40→1 scale, so mid-grid merit is measured rather than rounded to zero. The Drivers' title is complemented by an advisory **Driver-Merit Index**: a hierarchical Bayesian model that separates a car effect from a driver effect using teammate comparisons (identical machinery), tied onto one scale and reproducible from on-chain data — revealing the best driver net of machinery without corrupting team behaviour. A signature rule replaces blue flags with *lapping elimination*: a backmarker may fight, but if it is lapped by the leader it is black-flagged out, and the leader scores a point. Because only the leader can ever lap a car first, the rule cleanly removes lapped-traffic interference — every car still running is on the lead lap.

8 Safety: the invariant

Safety is the one part of the formula that is never an objective to be optimised and never traded: a non-negotiable, outcome-based floor (crash-test pass criteria, a halo-equivalent cockpit structure, homologated energy storage, driver equipment, and circuit/medical/spectator-separation minima written at the ruleset level). Engineering freedom and rigid safety standards are orthogonal — one can run a tightly tuned competition while mandating exact crash-test outcomes.

9 Economic model: pinning the cap and the floor

The two keystone numbers were set from an explicit model rather than guessed. **Cap.** A bottom-up cost build for a lean, shared-design program puts a competent entry at ∼\$67 M and a running-cost floor at ∼\$56 M, so a **\$75 M** cap is ∼1.1× competent — a genuinely tight constraint with real development headroom, while still binding the teams who would outspend it. **Floor.** An innovate-versus-copy game shows that with no floor, teams self-select only ∼13% original-R&D (the public-goods under-provision that justifies the floor). Vitality rises monotonically with the floor at no cost to merit-correlation or field-compression, so the floor is a pure vitality-versus-affordability lever; and because the running-cost floor is ∼75% of the cap, the floor must sit above it to bite — hence **80% of the cap**.

10 Conclusion

Formula Zynerji is, in one line, the engineering freedom of 1970s Formula 1 wearing 2020s safety and a blockchain economy. By forcing disclosure, capping money, mandating a contribution floor, pricing the dirty-air externality, keeping the driving manual, and measuring driver merit explicitly, it makes the championship a faithful instrument for merit — and turns the sport's own shared R&D into the revenue that funds it.

The full regulations (technical, sporting, safety, financial), the mechanism catalogue, the economic model, and the figure sources are openly available at `github.com/Zynerji/FormulaZynerji`.

A Mechanism catalogue (M1–M13)

Every rule traces to a mechanism. The nine design razors (R1–R9) that generate them — e.g. *equalise inputs, never outputs*; *price externalities, don't prescribe around them*; *design for the equilibrium, assume Goodhart* — are set out in the project's `mechanism-design` document. "Removed" entries are retained for the record: they were superseded either by the self-balancing economy or by the move to conventional racing, and the table documents *why* each was dropped.

ID	Mechanism	Role	Status
M1	Money-neutral hard cap	Equal, hard \$75 M budget cap; no luxury-tax lane. Removes budget-bias.	Active
M2	Development-resource auction	Token market rationing wind-tunnel/CFD/dyno throughput.	Removed (self-balance)
M3	Merit-weighted handicap	Success-scaled cut to a team's *future* development.	Removed (self-balance)
M4	Dirty-air externality pricing	Wake is measured; a clean-wake car earns a larger design-stage downforce allowance. Enables merit overtaking → no DRS.	Active
M5	Luck-suppressing neutralisation	Gap-preserving / time-neutral safety-car correction.	Removed (conventional racing)
M6	Championship-estimator design	All-finishers convex points curve (40→1) tuned for signal-to-noise.	Active
M7	Driver-merit disentanglement	Advisory Driver-Merit Index: a Bayesian car-vs-driver split from teammate pairs.	Active
M8	Anti-gaming sporting rules	Sensor-measured track limits; broad parc-fermé; scrutineering by hash.	Active
M9	Robust cost monitoring	Related-party pricing, randomised audits.	Subsumed by M10
M10	**Forced on-chain disclosure**	Only on-chain parts are legal; the spine. Collapses spend; advantages go transient.	Active (spine)
M11	Innovation incentive	The reward to innovate: transient head-start + Innovation-Index recognition (no tokens).	Active
M12	**Minimum-spend floor**	Mandatory ≥80% of cap on development, ≥half original on-chain work; forecloses copy-only stagnation.	Active (keystone)
M13	Lapping elimination	No blue flags; lapped by the leader → black-flagged out, leader +1. Removes lapped-traffic luck.	Active

The active set is therefore: the blockchain spine (M10) and its two innovation safeguards (M11 carrot, M12 stick); a money cap (M1); dirty-air pricing (M4); and the competition estimators and rules (M6–M8, M13). The economy needs no artificial currency (M2/M3 gone), and output-side luck is left to conventional racing (M5 gone).

B Dimension & specification set

First-pass values; the era-kitbash basis and the derivation are in the project's `chassis-integration` document, the engine and systems in `technical.md`, and the cap/floor in `economic-modeling`.

B.1 Era basis

System	Era basis
Front & rear wings, floor, tyres, wheels, safety	2026
Suspension (hydraulically interconnected)	2021
Gearbox (7-speed + reverse)	2013
Chassis dimensions & body rules	2008
Engine, vectoring diff, economy	new

B.2 Dimensions & mass

Parameter	Value
Wheelbase	3300 mm
Overall length	≈4850 mm
Max width	1800 mm
Max height	950 mm
Front / rear track	≈1520 / 1425 mm
Wheels	18 in
Tyre dia. F / R	≈705 / 690 mm
Min. mass (car+driver, no fuel)	605 kg
Race fuel	≈80 kg / 100 L
Floor	2026 ground-effect, ≈0.92× scaled

B.3 Powertrain & economy

Parameter	Value
Engine	2.5 L I5 2-stroke uniflow diesel
Bore × stroke	≈86 × 86 mm (square)
Induction	screw (Lysholm) supercharger
Boost (derived)	PR ≈3.3 (≈2.2 bar gauge)
BMEP / rev ceiling	26 bar / 7000 rpm
Peak power	≈758 kW (≈1015 hp)
Peak torque	≈1155 N m @ 4200 rpm
Power-to-weight	≈1.25 kW/kg
BSFC / piston speed	≈190 g/kWh / ≈20 m/s
Injection	common-rail DI, ≈2500–3000 bar
Exhaust	electronic VLEM (set-and-run)
Fuel	JP-8-class synthetic e-kerosene
Budget cap / floor	$75 M / 80% of cap

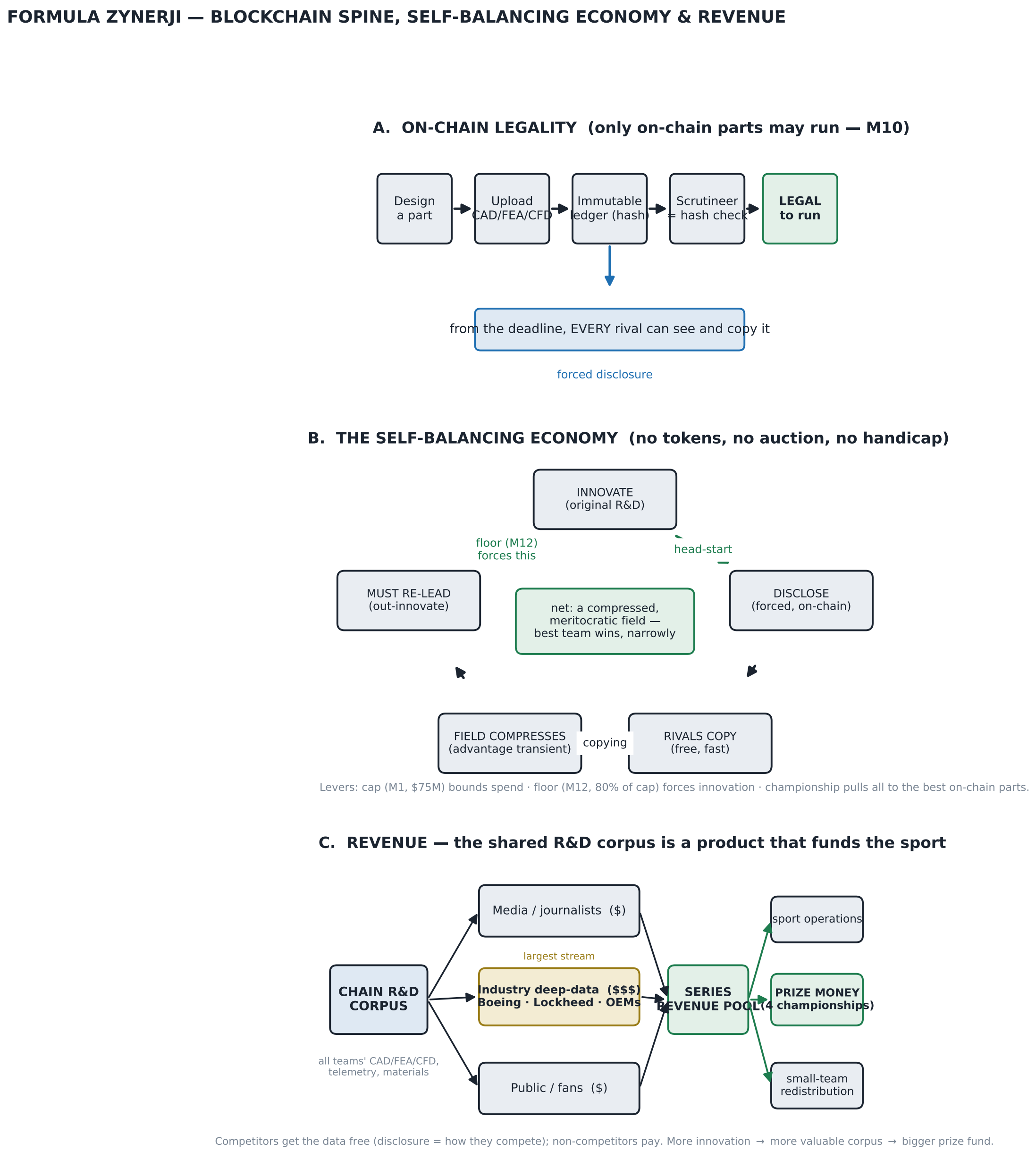

Figure 1: The blockchain spine: on-chain legality (A), the self-balancing economy (B), and the revenue streams that fund the sport (C).

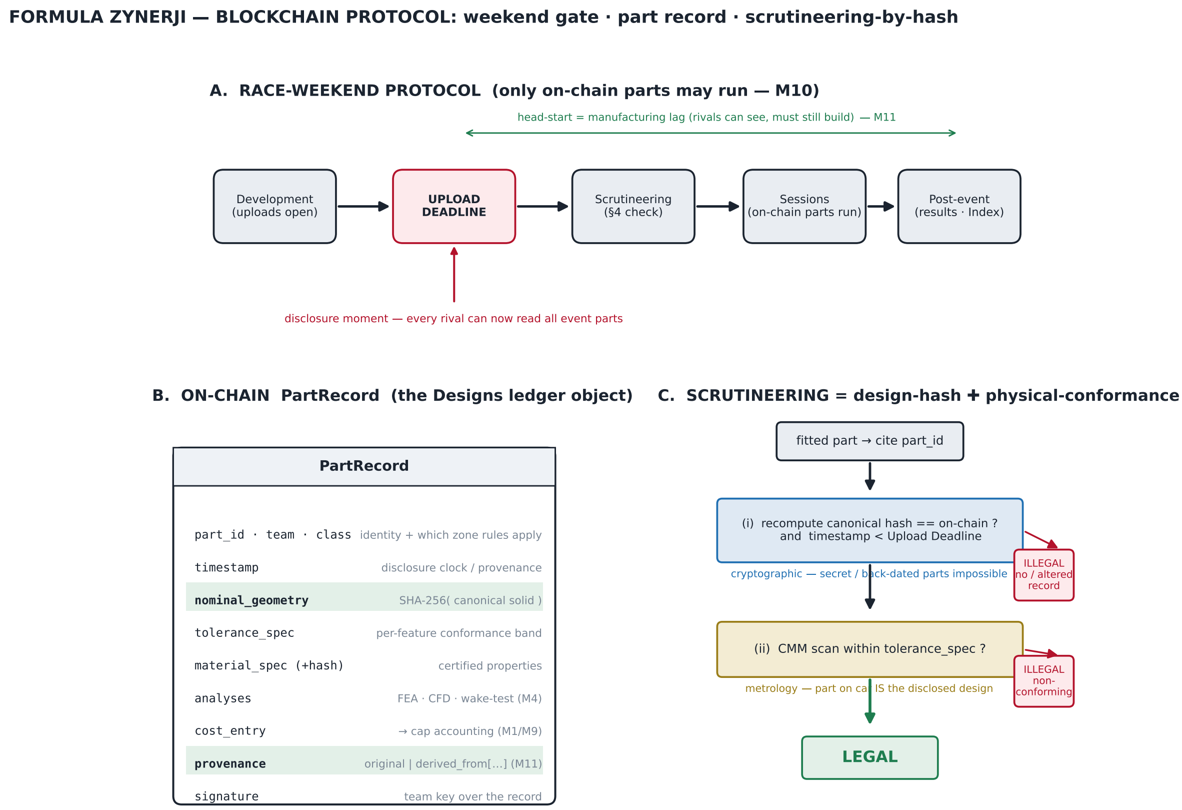

Figure 2: The spine's mechanics: the race-weekend gate (A), the on-chain `PartRecord` schema (B), and two-clause scrutineering — design-hash plus physical-conformance (C).

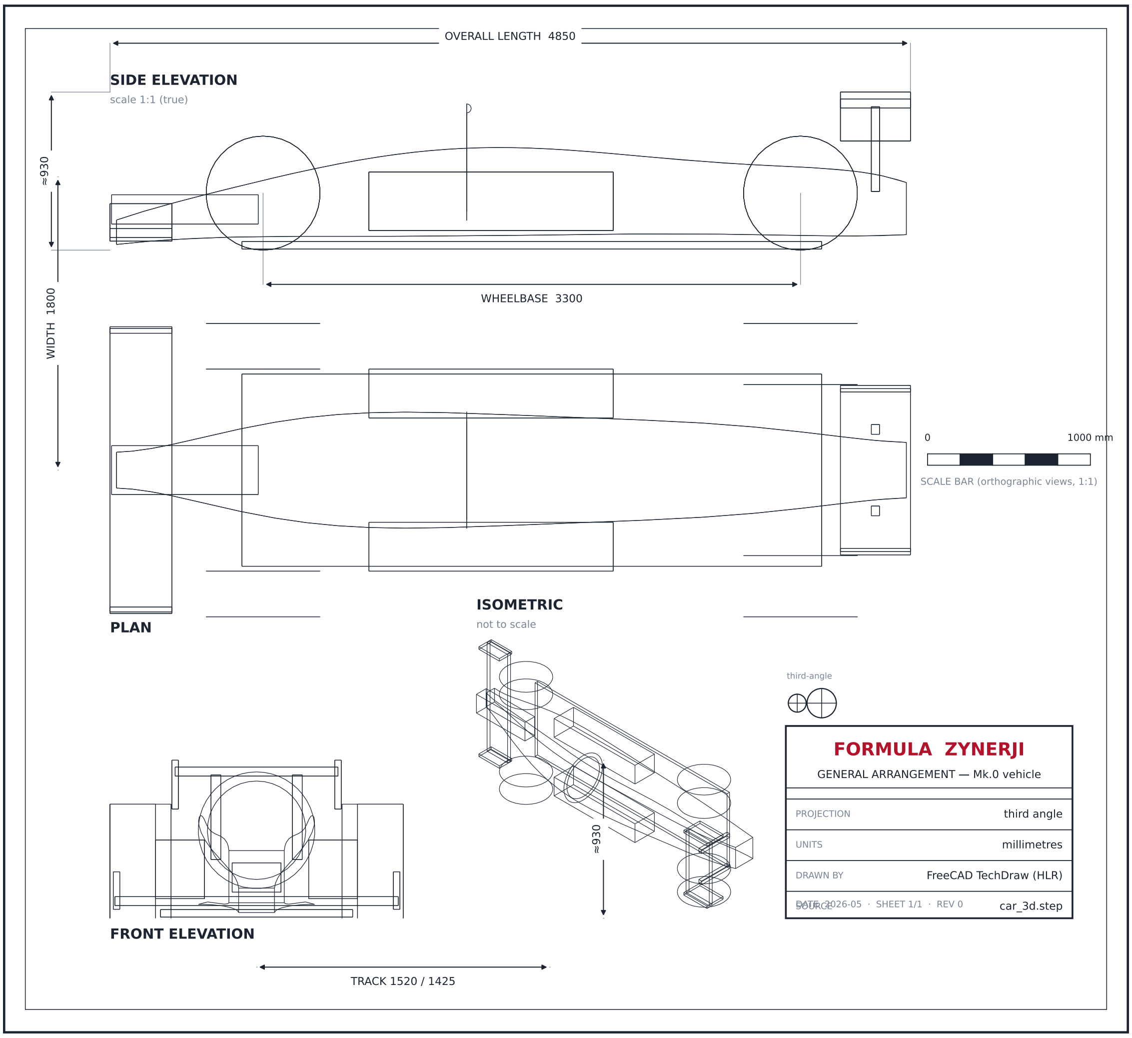

Figure 3: General arrangement — third-angle orthographic trio (plan, side, front) at a true shared scale plus a fit-to-box isometric, generated by hidden-line removal (FreeCAD TechDraw) directly from the parametric `car_3d.step` solid.

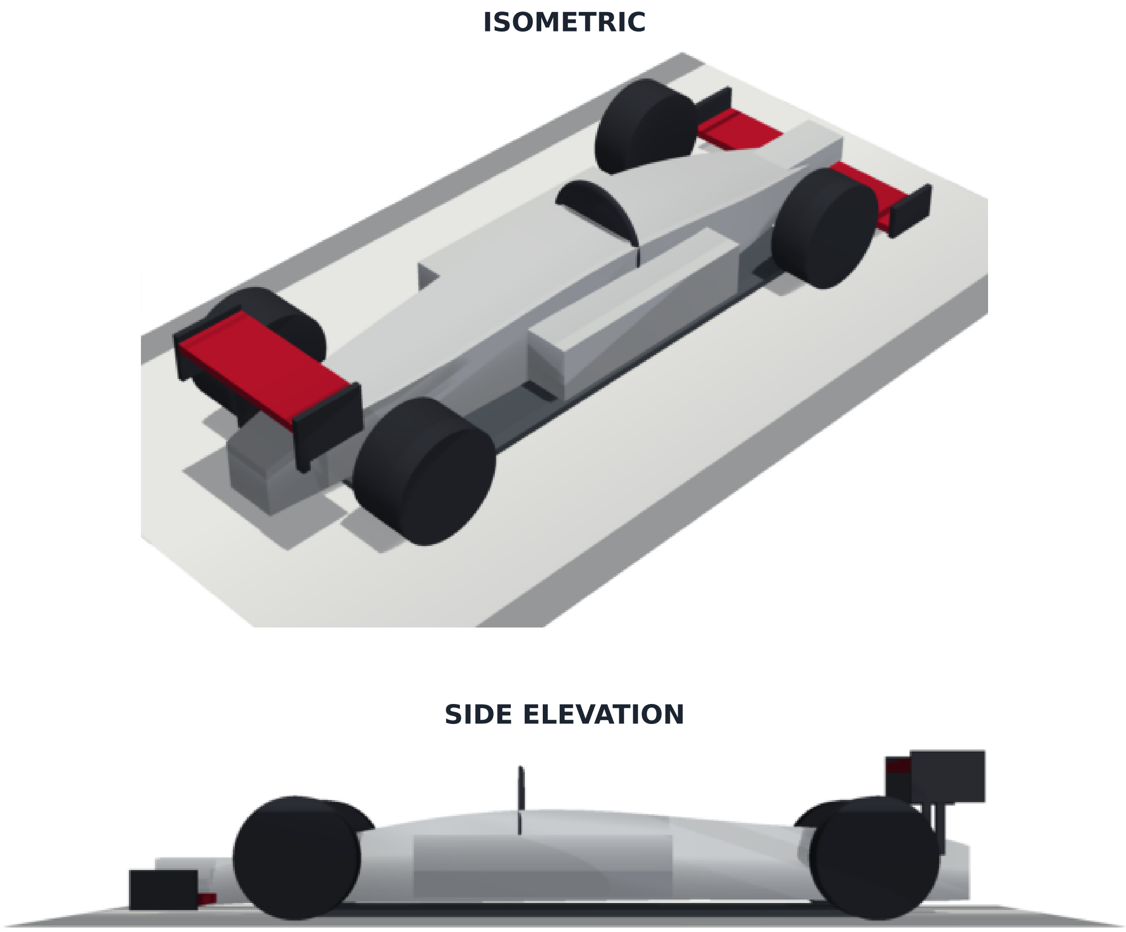

Figure 4: The same parametric solid as a lit 3D render (CadQuery model → STEP; PyVista).

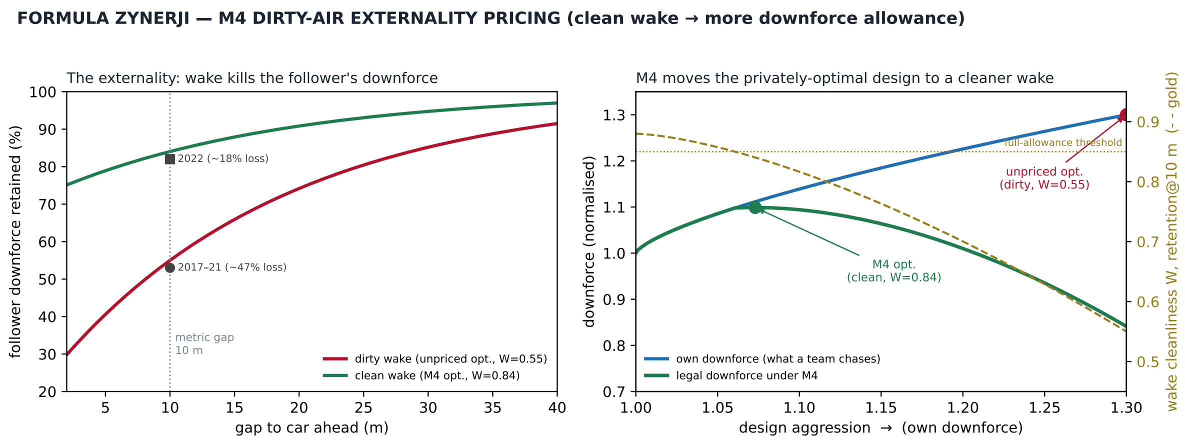

Figure 5: M4 dirty-air pricing made concrete. Left: a wake destroys the following car's downforce (the externality being priced). Right: scaling the legal downforce allowance by wake cleanliness moves the privately-optimal design to a far cleaner wake.

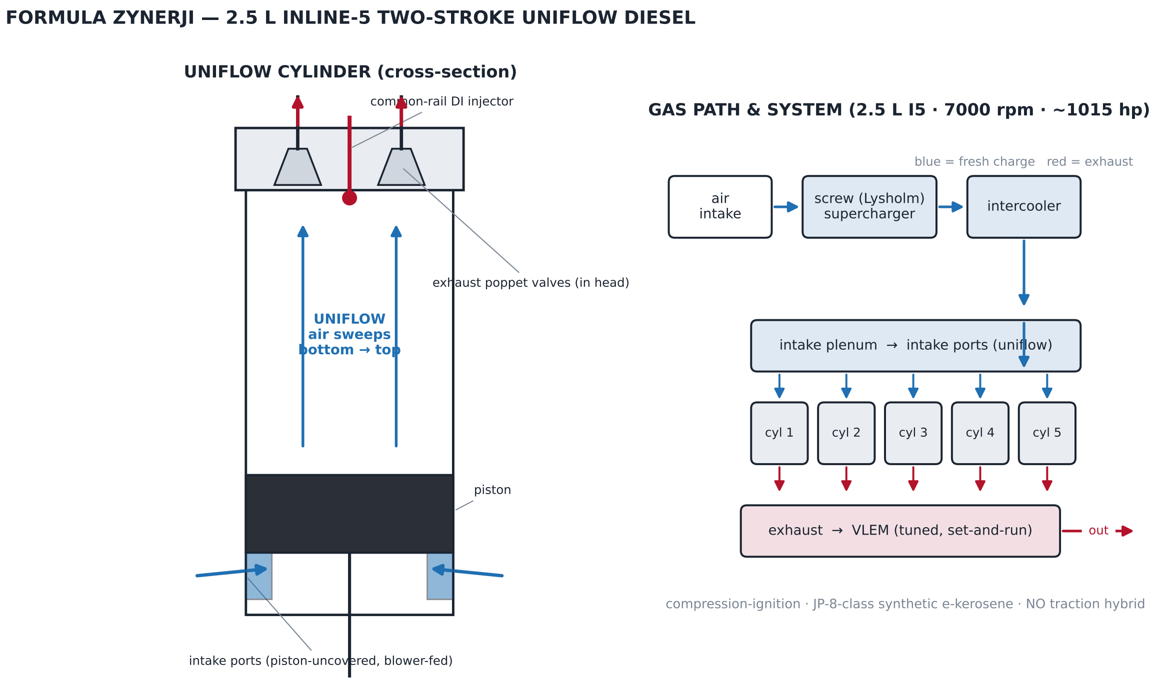

Figure 6: The 2.5 L I5 two-stroke uniflow diesel: cylinder cross-section and gas-path.

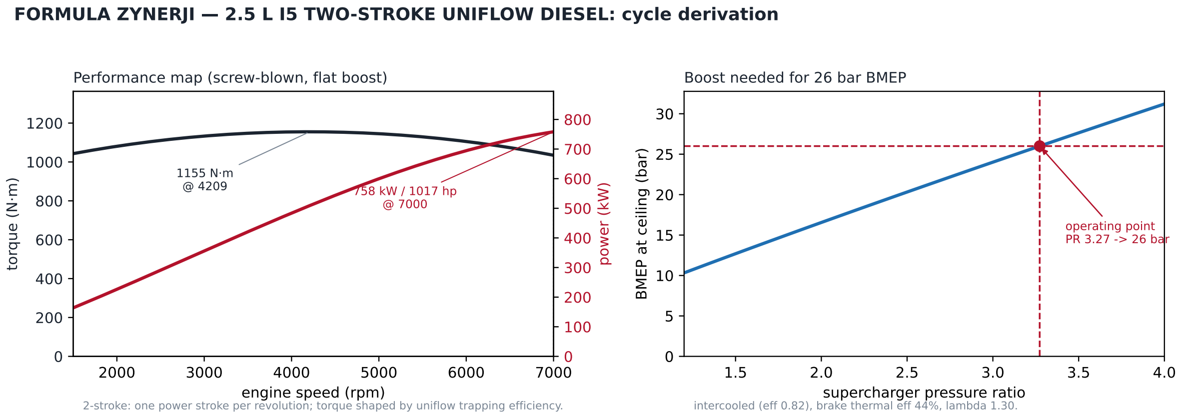

Figure 7: First-principles cycle derivation: the screw-blower boost required for 26 bar BMEP (right), and the resulting torque/power map (left) — peak torque ∼1155 N m mid-range, peak power 758 kW (1017 hp) at the 7000 rpm ceiling.

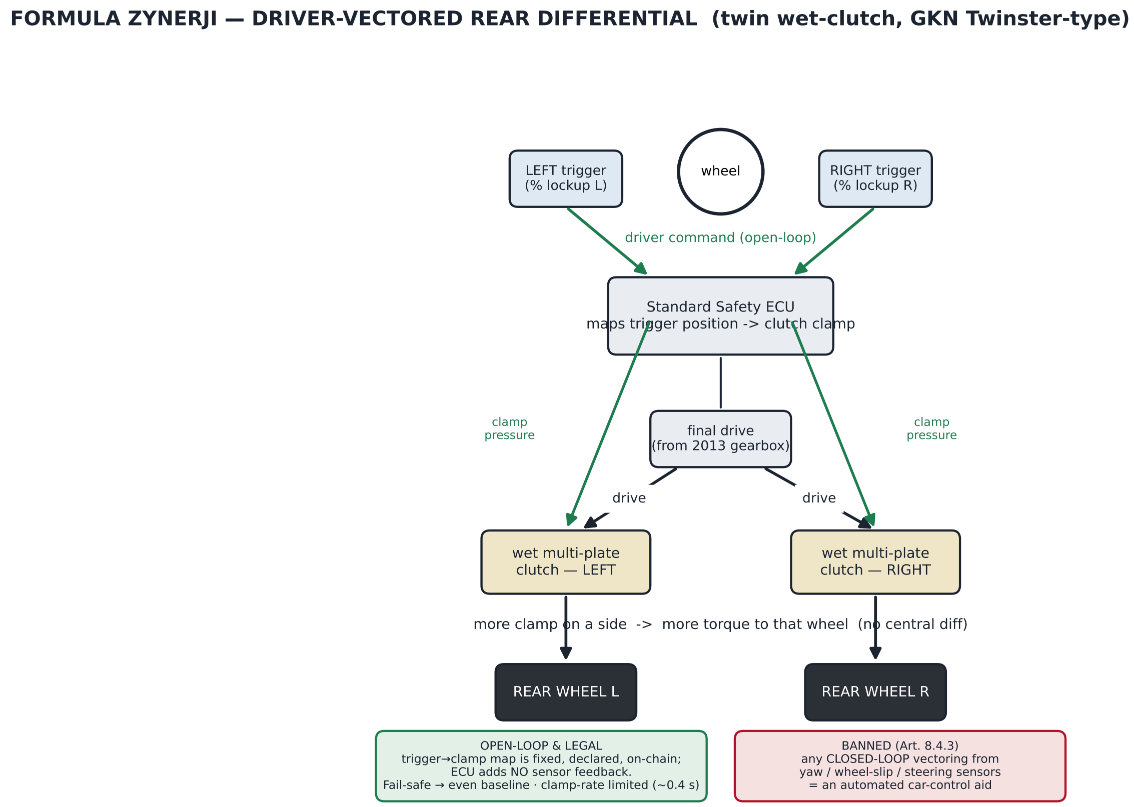

Figure 8: The driver-vectored twin wet-clutch differential and its open-loop control law.

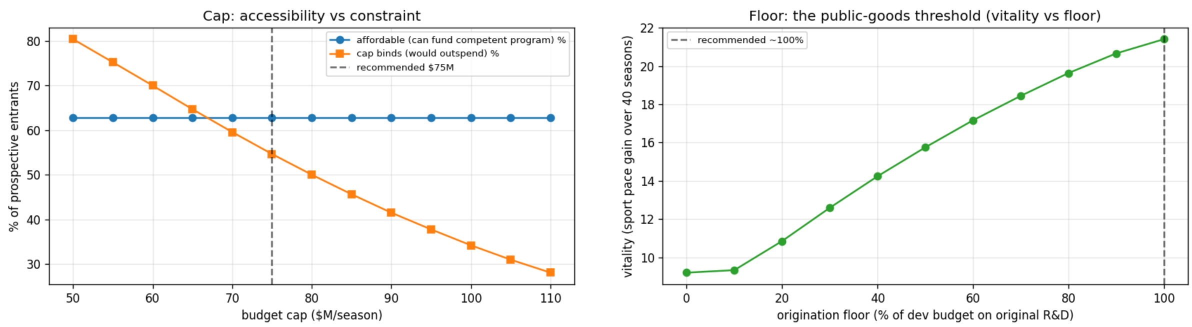

Figure 9: Left: cap accessibility versus constraint. Right: the floor as the public-goods threshold (vitality versus the origination floor).